

Experiment 6: SSL/TLS Vulnerability Analysis using sslscan

Aim

To analyze the SSL/TLS configuration of a website/server using sslscan, in order to detect weak protocols, weak ciphers, and certificate issues.

Requirements

A Kali Linux machine (Kali Linux recommended) with the required tool installed, and an SSL/TLS target machine on the same isolated lab network (e.g., Metasploit VM 2 / CMAA).

Algorithm / Procedure

1. Identify the target HTTPS server/IP in the lab environment.
2. Open a terminal on the Kali Linux machine.
3. Run sslscan against the target host and port (usually 443).
4. Review the list of supported SSL/TLS protocol versions (look for SSLv2/SSLv3/TLS1.0 which are weak).
5. Review the list of supported cipher suites (look for weak/export ciphers, e.g., RC4, DES).
6. Check the certificate details reported (validity period, key length, signature algorithm).
7. Document the weaknesses found and the recommended secure configuration (TLS 1.2/1.3 only, strong ciphers).

Sample Command / Code

```
# Basic sslscan against a target
sslscan 192.168.56.101:443

# Scan and save results to a file
sslscan --show-certificate 192.168.56.101:443 > sslscan_report.txt

# Alternative tool for comparison: testssl.sh
testssl.sh https://192.168.56.101
```

Result

The above procedure was successfully executed in the lab environment and the observed output was recorded/verified. (Attach screenshot of your terminal output here.)

Output :-

SSLscan testfire.net

Version : 2.0.11 - Static

OpenSSL 1.1.1n-dev xx xxx xxx.

connected to 65.61.137.117.

Testing SSL server testfire.net
on port 443 using SNI name
testfire.net.

SSL/TLS protocols:

SSLV2 disabled

SSLV3 disabled

TLSV1.0 enabled

TLSV1.1 disabled

TLSV1.2 enabled

TLSV1.3 disabled.

TLS Fallback SCSV:

Server supports TLS Fallback SCSV.

TLS renegotiation:

Session renegotiation got supported

TLS compression:

compression disabled.

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